

Q1. What is deep learning and how is it different from traditional machine learning?

Answer: Deep learning is a subset of machine learning that uses artificial neural networks with many layers to automatically learn complex patterns from large amounts of data. Unlike traditional machine learning, which requires manual feature engineering, deep learning learns features directly from raw data such as images, audio, or text, making it much more powerful for complex tasks.

Q2. What is an Artificial Neural Network (ANN) and how is it structured?

Answer: An Artificial Neural Network is a computational model inspired by the human brain, consisting of layers of interconnected nodes called neurons. It has three main parts: an input layer that receives data, one or more hidden layers that learn patterns, and an output layer that produces predictions. Each neuron processes input, applies a weight, adds a bias, and passes the result through an activation function.

Q3. What is an activation function and why is it important in a neural network?

Answer: An activation function is a mathematical function applied to the output of each neuron to introduce non-linearity into the network. Without it, the network would behave like a simple linear model regardless of how many layers it has. Common activation functions include ReLU, Sigmoid, and Tanh, each controlling how and whether a neuron fires based on its input.

Q4. What is the difference between ReLU, Sigmoid, and Tanh activation functions?

Answer: Sigmoid squashes output between 0 and 1, making it useful for binary classification but prone to vanishing gradients. Tanh squashes output between -1 and 1 and is zero-centered, making it better than Sigmoid in many cases but still prone to vanishing gradients. ReLU outputs the input directly if positive and zero otherwise, making it faster and less prone to vanishing gradients, so it is the most widely used in deep networks.

Q5. What is forward propagation in a neural network?

Answer: Forward propagation is the process where input data passes through the network layer by layer from the input to the output. At each layer, the input is multiplied by the weights, a bias is added, and the result is passed through an activation function. The final output is compared with the true label to compute the loss, which measures how wrong the prediction is.

Q6. What is backpropagation and how does it train a neural network?

Answer: Backpropagation is the algorithm used to train neural networks by computing the gradient of the loss with respect to each weight in the network. It works by applying the chain rule of calculus to propagate the error backward from the output layer to the input layer. These gradients are then used by an optimizer to update the weights in a direction that reduces the loss.

Q7. What is gradient descent and what are its main variants?

Answer: Gradient descent is an optimization algorithm that updates model weights by moving in the direction that reduces the loss function. The three main variants are batch gradient descent, which uses the entire dataset for each update; stochastic gradient descent (SGD), which uses one sample at a time; and mini-batch gradient descent, which uses a small batch of samples, balancing speed and stability.

Q8. What is the vanishing gradient problem and how can it be solved?

Answer: The vanishing gradient problem occurs when gradients become extremely small as they are propagated backward through many layers, making the earlier layers learn very slowly or not at all. It is especially common with Sigmoid and Tanh activations. Solutions include using ReLU activation, applying batch normalization, using residual connections (as in ResNet), or using better weight initialization methods like Xavier or He initialization.

Q9. What is dropout regularization and how does it prevent overfitting in deep learning?

Answer: Dropout is a regularization technique where a random fraction of neurons are temporarily ignored (set to zero) during each training step. This forces the network to not rely on any single neuron and learn more robust and distributed representations. At inference time, all neurons are active but their outputs are scaled accordingly, making the model generalize better to unseen data.

Q10. What is batch normalization and why is it used in deep neural networks?

Answer: Batch normalization is a technique that normalizes the output of each layer across a mini-batch, keeping the inputs to the next layer with a stable mean and variance. It speeds up training, reduces sensitivity to weight initialization, and acts as a mild regularizer. It is applied between the linear transformation and the activation function in most modern deep networks.

Q11. What is a Convolutional Neural Network (CNN) and what makes it suited for images?

Answer: A Convolutional Neural Network is a type of deep neural network specifically designed for processing grid-like data such as images. It uses convolutional layers that apply learnable filters across the input to detect local features like edges, textures, and shapes. Because filters are shared across the entire image, CNNs require far fewer parameters than fully connected networks and naturally capture spatial relationships in the data.

Q12. What is a convolution operation in a CNN and what does a filter do?

Answer: A convolution operation slides a small matrix called a filter or kernel over the input image and computes the dot product between the filter and the overlapping patch of the image

at each position. This produces a feature map that highlights regions where the learned pattern appears. Different filters detect different features, such as edges in one direction or color gradients.

Q13. What is pooling in a CNN and what are the types of pooling?

Answer: Pooling is a down-sampling operation that reduces the spatial size of feature maps to decrease computation and create translation invariance. Max pooling takes the maximum value from each region, retaining the strongest activation. Average pooling takes the average value. Max pooling is more commonly used because it preserves the most prominent features while discarding less important information.

Q14. What is the role of the fully connected layer in a CNN?

Answer: After several convolutional and pooling layers extract spatial features from the image, the output is flattened into a one-dimensional vector and passed through one or more fully connected (dense) layers. These layers combine the learned features to make final predictions, similar to a standard artificial neural network. The last fully connected layer typically uses Softmax for multi-class classification.

Q15. What is padding in a CNN and why is it used?

Answer: Padding refers to adding extra border pixels (usually zeros) around the edges of the input before applying a convolution. Without padding, the spatial size of the feature map shrinks with each convolutional layer, and edge pixels contribute less to the output. Padding preserves the original input size and ensures that all regions, including edges, contribute equally to the feature extraction.

Q16. What are popular CNN architectures and what made each significant?

Answer: LeNet was one of the earliest CNNs designed for digit recognition. AlexNet won the 2012 ImageNet challenge and proved deep CNNs work at scale. VGGNet showed that deeper networks with small filters improve accuracy. GoogLeNet introduced inception modules for efficiency. ResNet introduced residual connections to train very deep networks without vanishing gradients, enabling architectures with hundreds of layers.

Q17. What is transfer learning in the context of CNNs?

Answer: Transfer learning is the practice of using a CNN pre-trained on a large dataset like ImageNet as a starting point for a new task with less data. The earlier layers, which capture general features like edges and textures, are reused, while the later layers are retrained or fine-tuned for the new task. This drastically reduces training time and data requirements while achieving high accuracy.

Q18. What is a Recurrent Neural Network (RNN) and what problem does it solve?

Answer: A Recurrent Neural Network is a type of neural network designed for sequential data such as text, speech, or time series. Unlike feedforward networks, RNNs have connections that loop back to previous time steps, allowing them to maintain a hidden state that acts as memory. This makes them capable of capturing temporal dependencies and context across a sequence.

Q19. What is an LSTM and how does it solve the vanishing gradient problem?

Answer: Long Short-Term Memory (LSTM) is a special type of RNN with a more complex cell structure containing three gates: an input gate, a forget gate, and an output gate. These gates control what information is added to, removed from, or read from a dedicated cell state. The cell state acts as a long-term memory that can carry information across many time steps without the gradient shrinking, solving the vanishing gradient issue.

Q20. What is the difference between a one-to-many, many-to-one, and many-to-many RNN architecture?

Answer: One-to-many takes a single input and produces a sequence, such as image captioning. Many-to-one takes a sequence and produces a single output, such as sentiment analysis. Many-to-many takes a sequence and produces a sequence of the same or different length, such as machine translation or video labeling. These different architectures allow RNNs to handle a wide variety of sequential tasks.

Q21. What is the attention mechanism and why was it introduced for sequence models?

Answer: The attention mechanism was introduced to allow models to focus on the most relevant parts of the input when producing each output, rather than compressing the entire input into a single fixed-size vector. In sequence-to-sequence models, attention computes a weighted sum of all encoder states for each decoder step, giving the model the ability to look back at any part of the input and dramatically improving performance on long sequences.

Q22. What is the Transformer model and how is it different from RNNs?

Answer: The Transformer is a deep learning architecture that relies entirely on self-attention mechanisms instead of recurrence to process sequences. Unlike RNNs, which process tokens one at a time sequentially, Transformers process all tokens in parallel, making them much faster to train. Self-attention allows every token to directly attend to every other token in the sequence, capturing long-range dependencies more effectively than RNNs.

Q23. What is self-attention in a Transformer and how does it work?

Answer: Self-attention is a mechanism where each token in a sequence computes a weighted relationship with every other token in the same sequence. For each token, three vectors are computed — query, key, and value — using learned weight matrices. The dot product of the

query with all keys determines attention scores, which after softmax become weights applied to the values to produce a context-aware representation for each token.

Q24. What is Computer Vision (CV) and what are its major tasks?

Answer: Computer Vision is a field of AI that enables machines to interpret and understand visual information from the world, such as images and videos. Its major tasks include image classification (assigning a label to an image), object detection (finding and locating objects with bounding boxes), image segmentation (classifying every pixel), face recognition, pose estimation, and optical character recognition (OCR).

Q25. What is the difference between image classification, object detection, and image segmentation?

Answer: Image classification assigns a single label to an entire image. Object detection identifies multiple objects within an image and draws bounding boxes around each one along with their class labels. Image segmentation goes further and classifies every individual pixel in the image. Semantic segmentation assigns a class to each pixel while instance segmentation additionally distinguishes separate instances of the same class.

Q26. What is YOLO and why is it popular for object detection?

Answer: YOLO (You Only Look Once) is a real-time object detection algorithm that treats detection as a single regression problem, predicting bounding boxes and class probabilities in one pass through the network. Unlike older two-stage detectors like R-CNN, YOLO is extremely fast because it processes the entire image in a single forward pass. This makes it ideal for applications requiring real-time detection, such as autonomous driving or surveillance.

Q27. What are image data augmentation techniques and why are they used in CV?

Answer: Image data augmentation involves applying random transformations to training images to artificially increase the size and diversity of the dataset. Common techniques include flipping, rotation, cropping, brightness adjustment, zoom, and adding noise. Augmentation helps prevent overfitting by ensuring the model does not memorize specific training examples and learns features that generalize to unseen variations of the input.

Q28. What is Natural Language Processing (NLP) and what are its core tasks?

Answer: Natural Language Processing is a field of AI that enables computers to understand, interpret, and generate human language. Core tasks include text classification, sentiment analysis, machine translation, named entity recognition, question answering, text summarization, and language generation. NLP uses statistical and deep learning methods to process raw text into meaningful representations that machines can work with.

Q29. What is word embedding and why is it preferred over one-hot encoding for text?

Answer: Word embedding is a technique that represents words as dense, low-dimensional real-valued vectors where semantically similar words are placed close together in the vector space. One-hot encoding represents each word as a sparse vector with a single 1, which is high-dimensional and captures no semantic relationships. Embeddings like Word2Vec and GloVe learn meaningful representations where relationships like king minus man plus woman equals queen are captured numerically.

Q30. What is sentiment analysis and how is it done using deep learning?

Answer: Sentiment analysis is the task of identifying the emotional tone or opinion expressed in a piece of text, such as classifying a review as positive, negative, or neutral. Using deep learning, text is first converted into embeddings, then passed through layers such as LSTMs, CNNs, or Transformer encoders to extract context-aware features, and finally a fully connected layer with Softmax outputs the sentiment class. Pre-trained models like BERT can be fine-tuned for this task with high accuracy.